

Assignment 01

TG/2022/1388

Question 01

A). Saman is a citizen who lives in smart urban city he wants to pay his electricity bills. Then he opens the Smart Urban Service app on their phone and log in and select “Utility Payments” to settle his bills. He can see his bill amount and can pay it online. And also check the bus timing and can report garbage collection issues quickly. This system helps citizens to do many services in one place easily without wasting time.

B). Concrete scenario- paying a utility bill (Electricity like)

User- Saman

Goal-To pay a monthly water bill immediately

Steps-

1. Saman opens the smart urban service app on his smartphone or etc.
2. The smart phone connects to the internet (using wifi or cellular).
3. Logs into his account using correct credential (username & password) or biometrics.
4. After go to utility payment section.
5. System gets the correct details from government database server
6. And display amount of bill on a smart screen.
7. User can select payment methods (card or mobile wallet)
8. The system confirms payment and sends successful notifications.

Question 02

1. People (Users)

There are different types of users engage with system.

Working professionals -they need fast & correct service

Elderly people- They need simple and easy UI interface for working

Tourists – They need different language support

People with disabilities- They need voice or accessibility features to work with system

Government staff -They manage and monitor system easily

2. Activities

Pay bills, check the public transport, report problems quickly, schedule garbage collection & emergency reporting.

Some activities can be simple and some's activities can be complex because of the internal activities are big.

3. Context

Users use this system at home, on the road, in office and low internet areas and noisy, and high internet speed access areas etc.

4. Technologies

Users use this system using mobile apps, web applications, kiosks, voice assistant, internet and cloud services.

Question 03

A)

Transport loop:

- users check the bus timetable or location
- The system provides time and gps location
- and users can decide what to do

waste loop:

- users report a full the garbage bin via application
- system can optimize the truck route

payment loop:

- system sends bill notification for user
- users pay the bill via mobile wallet
- system can update balance

B)

Technology can change user activity quickly:

- Before technology grow: People go to offices to do their jobs
- Now: They use mobile apps to do their work

User needs influence technology for work:

- Elderly people - need big buttons to be visible clearly
- Disabled users - need voice control etc

Question 04

1. People + Technology

- Problem: Elderly people can not understand complex apps and UI designs
- Solution: Simple design, large text to clearly visibility

2. Activities + Context

- Problem: Reporting emergency cases in noisy place
- Solution: Use simple buttons instead of typing etc.

3. Technology + Context

- Problem: Low internet access areas
- Solution: Offline features or emergency hotline services (like 1919)

4. People + Activities

- Problem: Tourists can not understand local languages
- Solution: Multi-language support application need

5. Design implications:

- System must be simple UI interfaces
- Need work in every where
- support different users correctly

Question 05

A)

1. Complex ui interface

Problem: many options confuse users to their work correctly

PACT elements: People + Technology

Solution: Simple layout (UI designs)

Reason: Easier for all users to work clearly

2. Poor internet connectivity

Problem: System may not work in some areas (low internet access areas)

PACT elements: Context + Technology

Solution: Offline mode is better

Reason: Works everywhere

3. Accessibility issues

Problem: Disabled users cannot use system coreectly

PACT elements: People + Technology

Solution: Voice commands, screen readers need to implemet

Reason: Inclusive design

4. Slow system response

Problem: Takes too long to load

PACT elements: Technology + Activities

Solution: Faster servers

Reason: Saves time